

DESIGN CRITERIA

	Heating	Cooling
Room set point	16.0 °C	26.0 °C
Outdoor dry bulb	-4.9 °C	28.4 °C
Outdoor RH (coincident)	90 %	40 %
ROOM HUMIDITY		
Humidification set point	—	
lower RH limit		
Dehumidification set point	—	
upper RH limit		

SIZING (TAS DESIGN LOADS)

	Heating	Cooling
Design load	1,368 W	0 W
Design load per area	263.1 W/m²	0.0 W/m²
Sizing multiplier	not set	not set
Design loads: from Tas sizing — date/currency not recorded		

HEATING

Design day	London_Heathrow_Illustrative_Heating_Design_Day_99.6%_Coincident_Wet_Bulb_Extreme_Winter_Condition	
	Design day	Full year
Peak load	23.46 kW	15.43 kW
Peak hour	06:00–07:00	31 Dec 23:00–24:00
Room dry bulb	18.2 °C	18.0 °C
Room resultant	16.9 °C	17.1 °C
Room RH	23 %	45 %
Room humidity ratio	2.5 g/kg	5.1 g/kg
Outdoor dry bulb	—	-6.8 °C
Outdoor RH	—	95 %

SENSIBLE COMPONENTS AT PEAK

Component	Design day (kW)	Full year (kW)
Lighting	0.00	0.35
Occupancy (sensible)	0.00	0.42
Equipment (sensible)	0.00	1.25
Infiltration / ventilation	-8.12	-6.01
Air movement (inter-zone)	-1.23	-0.99
Building heat transfer	-3.46	-2.35
External conduction — opaque	-6.79	-5.12
External conduction — glazing	-3.85	-2.98

LATENT COMPONENTS AT PEAK

Component	Design day (kW)	Full year (kW)
Occupancy (latent)	0.00	0.31
Equipment (latent)	0.00	0.05

Zero at both peaks (not listed): Solar, Air handling unit
Components as the simulation reports them at each peak hour: + gain to the room air, – loss from it. No total is derived.

COOLING

Design day

London_Heathrow_Illustrative_Cooling_Design_Day_0.4%_Dry_Bulb=>Mean_Coincident_Wet_Bulb_With_Global_Radiation

	Design day	Full year
Peak load	31.23 kW	28.77 kW
Peak hour	15:00–16:00	27 Jul 15:00–16:00
Room dry bulb	24.0 °C	24.0 °C
Room resultant	25.8 °C	25.2 °C
Room RH	55 %	52 %
Room humidity ratio	10.2 g/kg	9.8 g/kg
Outdoor dry bulb	—	30.1 °C
Outdoor RH	—	38 %

SENSIBLE COMPONENTS AT PEAK

Component	Design day (kW)	Full year (kW)
Solar	12.35	11.23
Lighting	2.10	2.10
Occupancy (sensible)	3.15	3.15
Equipment (sensible)	7.50	7.50
Infiltration / ventilation	1.23	0.99
Air movement (inter-zone)	0.46	0.35
Building heat transfer	1.99	1.46
External conduction — opaque	2.35	1.88
External conduction — glazing	1.23	1.11
Air handling unit	-1.12	-0.99

LATENT COMPONENTS AT PEAK

Component	Design day (kW)	Full year (kW)
Occupancy (latent)	2.30	2.30
Equipment (latent)	0.15	0.15

Components as the simulation reports them at each peak hour: + gain to the room air, – loss from it. No total is derived.

RESULTS

	Heating	Cooling
Status	Available	Available
Result source	Tas	Tas
Read into model	27 Aug 2026 18:38	27 Aug 2026 18:38
Matches current model	not recorded	not recorded

Peak hour: a design-day peak has an hour of the day only, no date; a full-year peak shows day and hour, no year.
Design loads (sizing) and simulated peaks are shown side by side for information; no acceptance rule is applied.